2025 Physics Prelim P3 Solutions

1 **(a)** **(i)** $v_y^{\ 2} = u_y^{\ 2} + 2as_y$ M1

$= 0 + 2(9.81)(32)$

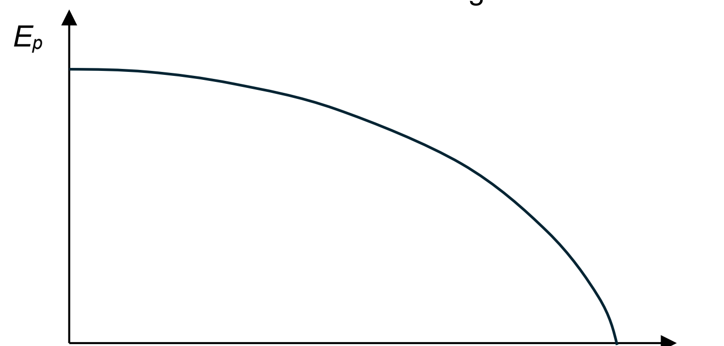
$v_y = 25.1 \text{ m s}^{-1}$ A1

(ii) $\sin\theta = \dfrac{25.1}{34}$ M1

$\theta = 47.6^{\circ}$ A1

(b) With splashing, there is a transfer of KE of stone to KE of water.
Less KE of stone available to do work against resistive forces in the water. B1

(c) **(i)**

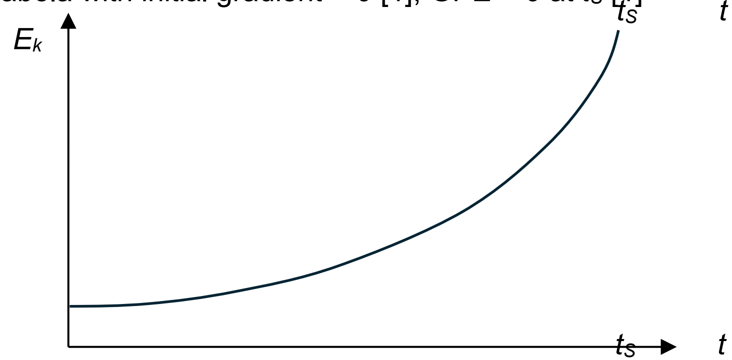


B1

B1

Parabola with initial gradient = 0 [1], GPE = 0 at t_S [1]

(ii)



B1